

Effect of Distance Between Trapped Ion Qubits on Decoherence Time in Trapped Ion Chains

Chenjia Ni

1 Introduction

Quantum computers offer a chance to revolutionize the computing scene and people's lives. These computers have the ability to surpass the capabilities of conventional classical computers particularly in fields such as chemistry, materials science, and finance [1]. The building blocks of these computers are qubits. Instead of the 0 and 1 states of a classical bit, qubits can be in any combination of $|0\rangle$ and $|1\rangle$ before it is measured and collapses to a single state. When it is in a combination of the $|0\rangle$ and $|1\rangle$ states, the qubit is in a superposition. Since these qubits can store more information than classical bits and can interact differently, quantum computers have an advantage over classical ones.

One of the main challenges to creating a universal quantum computer is coherence time, or how long superpositions and entanglement can be maintained within a system. Because of the fragility of quantum states and environmental noise, including thermal fluctuations and electromagnetic interference, usually coherence times are on the scale of microseconds [2]. Trapped ions, however, show incredible promise for creating qubits, as they have incredibly high decoherence time when compared to other forms of qubits, including photons, quantum dots, and superconducting circuits [2]. In these systems, ions are created, usually made from Be^+ , Mg^+ , Ca^+ , Ba^+ , or Yb^+ , and are levitated on a line using electric fields in Paul traps. The outermost valence electron and its hyperfine energy structure represents the quantum states, with a ground and excited state as $|0\rangle$ and $|1\rangle$ [3].

This research paper will analyze the effects of distance between the trapped ions on the coherence time in a simulated Yb^+ trapped ion quantum computer, as it has the required nuclear spin $1/2$, observational stability, first-order Zeeman insensitive clock states, and reasonable ionization energies. Recently, two industry quantum computers using ytterbium were introduced by Honeywell and IonQ [4]. We observe from our model that the coherence time depends on the distance between the ions, with the further apart the ions are, the less rapidly the ions decohere. We also observe that an increase in the number of ions in the chain increases the decoherence rate, however its effects on the co-

herence time are negligible compared to the distance between ions. In addition, our model did not predict any difference in coherence time based on the different type of ions used.

2 Model Hamiltonian

Numerous ways of modeling trapped ion systems exist that consist of two main terms, an energy Hamiltonian and an interaction Hamiltonian. We will follow the Hamiltonian given by N. Yazdanpanah and M. K. Tavassoly [5]. The authors used the following energy Hamiltonian to describe the system:

$$\hat{H}_0 = \hbar\nu(\hat{b}^\dagger\hat{b}) + \frac{1}{2}\hbar\omega_0\hat{\sigma}_z + \hbar\omega(\hat{c}^\dagger\hat{c} + \frac{1}{2}) \quad (1)$$

where ν is the frequency of the vibrational mode of phonon, $\hat{b}^\dagger\hat{b}$ is the annihilation (creation) operator of the phononic mode, and $\hat{c}^\dagger\hat{c}$ is the annihilation (creation) operator of the quantized field. Since we are assuming a low temperature model, the first term relating to the phononic mode (movement of atoms) can be set to 0. In addition, to simplify the model we will be ignoring the oscillations of the trapped ions within the Paul traps that cause the quantized field and will also set term 3 to 0. We are left with the second term which represents the internal energy state of the ion, where $\hbar$ is the reduced Planck's constant, ω_0 is the frequency of the internal ionic transition, and $\sigma_z^{(i)}$ is the pauli z-matrix for the i th ion. Since we model N different ions, we also need to sum together all the terms for each ion, leaving us with the following energy Hamiltonian:

$$\hat{H}_0 = \sum_{i=1}^N \frac{1}{2}\hbar\omega_0\sigma_z^{(i)} \quad (2)$$

In our case, Yb171+ has an internal ionic transition frequency of 12.642821 GHz which we set [4].

Next, we will find the interaction Hamiltonian. We will first consider the Hamiltonian given by N. Trautmann and P. Hauke [6]. The authors used the following interaction Hamiltonian to describe the system:

$$H_{\text{int}} = \sum_{i < j} J_{ij} \sigma_x^{(i)} \sigma_x^{(j)} \quad (3)$$

where $\sigma_x^{(i)}$ is the pauli x-matrix for the i th ion. However, they define the coupling strength J_{ij} as:

$$J_{ij} = \Omega_i \Omega_j \frac{\hbar k^2}{2m} \sum_n \frac{b_{i,n} b_{j,n}}{\Delta^2 - \nu_n^2}$$

where m is the ion mass, ν_n the eigenfrequencies of the transverse phononic modes, $b_{i,n}$ the elements of the normal-mode-matrix, k is the laser wavenumber, and Ω_i are the Rabi frequencies induced by the laser driving. In the energy

Hamiltonian, however, we ignored both phononic modes and the oscillations of the ions. In addition, this Hamiltonian equation does not account for the dipole-dipole interactions that arises from the fields of the ions in the trap.

In order to more accurately model for the dipole-dipole interactions, we next look at the Hamiltonian given by N. Yazdanpanah and M. K. Tavassoly, which includes the dipole-dipole interaction between the ions, given by the following equation for 2 dipoles:

$$H_{\text{int}} = \frac{1}{4\pi\epsilon_0|\vec{r}|^3} [\vec{Q}^{(1)} \cdot \vec{Q}^{(2)} - 3(\vec{Q}^{(1)} \cdot \hat{e}_r)(\vec{Q}^{(2)} \cdot \hat{e}_r)] \approx \frac{1}{4\pi\epsilon_0|\vec{r}|^3} \vec{Q}^{(1)} \cdot \vec{Q}^{(2)} \quad (4)$$

where ϵ_0 is the permeability of the vacuum, $\vec{Q}^{(i)}$ is the dipole moment of the i th dipole, $\vec{r}$ represents the relative position of the two dipoles, and $\hat{e}_r$ is the unit vector in that direction [5].

To create a simpler interaction Hamiltonian, we combine the structure from (3) with the r^3 term from (4) that characterizes the DDI interaction simplifying the Hamiltonian to:

$$H_{\text{int}} = \sum_{i < j} \frac{Q_{ij}}{r_{ij}^3} \sigma_x^{(i)} \sigma_x^{(j)} \quad (5)$$

Where $\sigma_x^{(i)}$ is the pauli x-matrix for the i th ion and Q_{ij} is a constant that represents the interaction strength between the i th and j th ion caused by the dipole moments ($\vec{Q}^{(1)} \cdot \vec{Q}^{(2)}$) chosen through experimental values [7].

Thus, our final Hamiltonian for the model is:

$$H = H_0 + H_{\text{int}} = \sum_{i=1}^N \frac{1}{2} \hbar \omega_0 \sigma_z^{(i)} + \sum_{i < j} \frac{Q_{ij}}{r_{ij}^3} \sigma_x^{(i)} \sigma_x^{(j)} \quad (6)$$

3 Model Lindblad Operator

Within the trapped ion system, the way ion states decay and dephase through environmental noise that can be modeled by the lindblad operator. In order to simulate noise within the system, we include the Lindblad operator given by N. Trautmann and P. Hauke [6]. Assuming the correlation time of the noise is 0, the dephasing processes that act independently on each site can be modeled by the Lindblad super operator:

$$\mathcal{L}_{\text{deph}}(\hat{\rho}) = \sum_i \frac{\gamma_i}{2} [-\{\hat{\sigma}_i^+ \hat{\sigma}_i^-, \hat{\rho}\} + 2\hat{\sigma}_i^+ \hat{\sigma}_i^- \hat{\rho} \hat{\sigma}_i^+ \hat{\sigma}_i^-] \quad (7)$$

with γ_i denoting the dephasing rate at site i , $\hat{\rho}$ is the density matrix of the system, and $\hat{\sigma}^+$ and $\hat{\sigma}^-$ are the raising and lowering operators for each qubit. In our case, a gamma dephasing rate of 10^{-4} was chosen [8].

4 Creating the QuTip Model

In order to model our system, we will be using QuTip, a python package for simulating the dynamics of open quantum systems. We begin by defining variables:

```

1  N = number_of_ions # Number of ions
2  h = 1.05457182e-34
3  omega_0 = 12.642821e9 # Qubit frequency
4  J = 1e-17 # Interaction strength
5  distance = distance_multiplier*1.0e-6
6  gamma_dephasing = 1e-4 # Dephasing rate
7
8  sx_list = [tensor([sigmax() if i == j else qeye(2) for i in
9  range(N)]) for j in range(N)]
10 sz_list = [tensor([sigmaz() if i == j else qeye(2) for i in
11 range(N)]) for j in range(N)]

```

The combined Hamiltonian using (6) is then calculated:

```

1  H = sum(h * omega_0 * sz_list[j] / 2 for j in range(N))
2  for i in range(N):
3      for j in range(i+1, N):
4          H += J / ((distance*(i-j))**3) * sx_list[i] * sx_list[j]

```

Next, all the ions are initialized in ground state

```

1  psi0 = tensor([basis(2, 0) for _ in range(N)])

```

The Lindblad operator from (7) is then constructed with the gamma dephasing constant:

```

1  c_ops = [np.sqrt(gamma_dephasing) * sz_list[j] for j in range(N)]

```

Finally, the system is solved for on a time scale of 10 seconds, and the coherence of the system is found:

```

1  t_list = np.linspace(0, 10, 100)
2  result = mesolve(H, psi0, t_list, c_ops, [])
3  coherence = [expect(sx_list[0], result.states[t]) for t in
4  range(len(result.states))]

```

5 Coherence of the Trapped Ion System

We now begin simulating our trapped ion system using Yb171+ as the ion. The smallest distance tested for coherence in our system was $4 \mu\text{m}$ as the current smallest trapped ion system was created with a separation of $4 \mu\text{m}$ [9]. Therefore, in figure 1, we compare the coherence times in a 4 ion system with a separation distance between the ions between $4 \mu\text{m}$ and $12 \mu\text{m}$

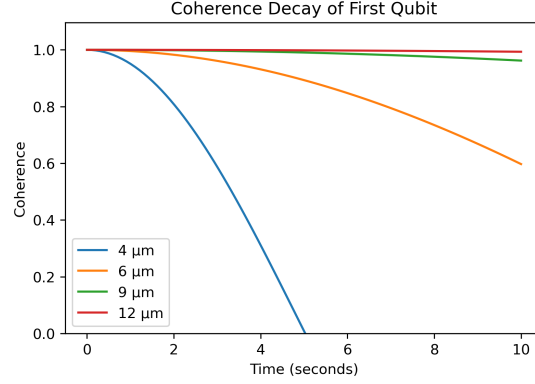


Fig. 1: Coherence over time for a 4 ion system with varying distances. Coherence values for the system with a 4 μm after it reached 0 was discarded as the coherence value oscillated after it reached 0.

In the 4 ion system, the coherence level decreased exponentially as a function of time, with the effect being more profound the shorter the distance between the ions were. After a distance of 9 μm , there was barely any change between the rate of coherence decay likely due to the cubed exponent on distance in the interaction Hamiltonian in (6).

An important benchmark for qubit performance is the fault-tolerance threshold, where a quantum error correction protocol can correct errors faster than the errors can occur. For our system, we use the benchmark of 0.993 coherence following a recent QEC protocol that achieves an error threshold of 0.7% [10]. In figure 2, we graphed the relationship between time over this coherence threshold and distance between ions.

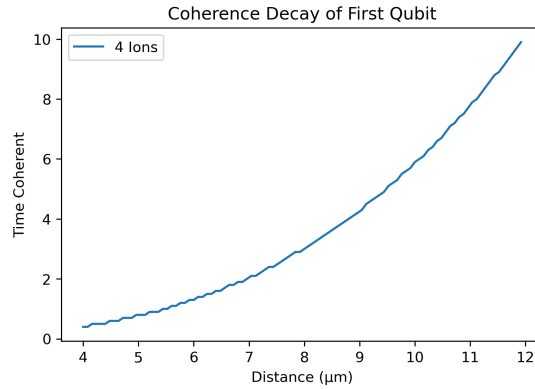
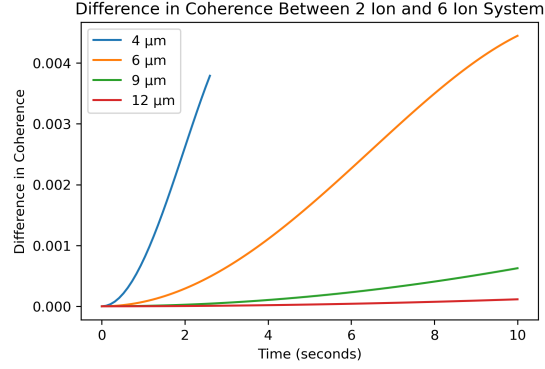


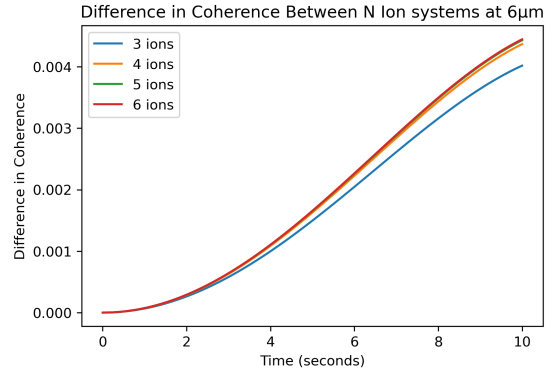
Fig. 2: Time spent over coherent threshold (0.993) for 4 ions with varying distances. The graph follows the equation $y = ax^b$ where $a = 0.00655862$ and $b = 2.95148$.

The amount of time the system remains above the coherence threshold increases exponentially along with distance, with the exponent $b = 2.95148$.

In figure 3, we find the effect of increasing numbers of ions on coherence time.



(a)



(b)

Fig. 3: Graph *a* depicts the difference in coherence between a 2 ion and a 6 ion systems (2 ion - 6 ion coherence). Similarly, differences in coherence values for the system with a $4 \mu\text{m}$ distance after it reached 2.6 seconds was discarded as the coherence value oscillated. Graph *b* depicts the difference in coherence between 2 ion and n ion systems (2 ion - n ion). The coherence was measured at ion spacing of $6 \mu\text{m}$

The difference in coherence between trapped ion systems with differing number of ions is marginal, on the scale of 10^{-4} . However, the systems with more ions decohere at a faster rate than systems with less ions. Figure 3 *a* shows how the effect of ions on coherence is more profound at smaller distances while at $12 \mu\text{m}$ the effect practically nonexistent. The effect of changing the number of ions while at a fixed distance between ions ($6 \mu\text{m}$) showed even less variability.

While the 3 ion system differed the least from the 2 ion system's decoherence, the differences were still on the scale of 10^{-4} , with barely any change between the 4,5, and 6 ion systems.

In addition, another commonly used ion, Ca^{43+} , was modeled with an internal ionic transition frequency of 3.2 GHz [11]:

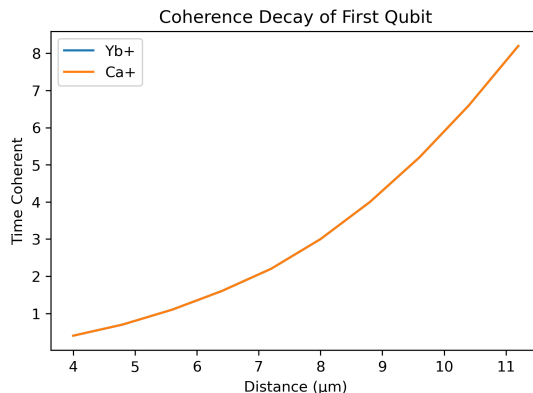


Fig. 4: Time spent over coherent threshold (0.993) between Ca^{43+} ions and Yb^{171+} ions.

The data did not show any difference at all in coherence between the trapped ion systems using different ions, even on the scale used for the difference in coherence between different numbers of ion systems. While this result suggests inaccuracy in our model Hamiltonian, it is important to note that the direct relationship between splitting size and coherence time is not always straightforward, as long coherence times have been demonstrated equally for differing ions like Ca^{43+} , Ca^{40+} , Be^{+} , and Yb^{171+} [4] [12] [13]. This suggests a more complicated relationship between coherence time and ion type, which could be more affected by the experimental setups and environment.

6 Conclusion

In summary, we have provided a simplified way to model a system of trapped ions and measure the coherence of the system over time. This approach uses software to simulate the evolution of these systems.

The data shows a large effect of distance between consecutive trapped ions on coherence, with an exponential relationship ($b = 2.95148$) between the distance between ions and time spent coherent (above 0.993). However, the number of trapped ions on the chain did not have that large of an effect on coherence from $n = 2$ to $n = 6$ ions. While more ions on the chain led to decreased coherence, the effect was negligible compared to the effect of distance. In addition, the model did not show any difference in coherence at all between the two ions Ca^{43+} and Yb^{171+} despite their different hyperfine energy gap. This could

be caused by the simplified Hamiltonian of the model which only includes the frequency in the energy Hamiltonian.

Starting with the model, many improvements exist with the Hamiltonian, whether it be adding back the terms related to the phononic modes and the oscillations of the ions used in the energy Hamiltonian or adding more nuance to the interaction Hamiltonian. These inaccuracies could have resulted in the strange behavior of coherence in the 4 μm simulations and the lack of difference in coherence between different ions. A more accurate interaction Hamiltonian would likely make the simulated effect of adding more ions on the chain more pronounced as well.

To conclude, the trapped ion design is a promising way forward to fully realized quantum computers, with its incredibly long coherence time. However, while adding more ions onto the chains have little effect on coherence, reducing the spacing between the ions serve as a limit to how compact and efficient this design can achieve.

7 References

- [1] Chae, E., Choi, J. and Kim, J. An elementary review on basic principles and developments of qubits for quantum computing. *Nano Convergence* 11, 11 (2024). <https://doi.org/10.1186/s40580-024-00418-5>
- [2] T. D. Ladd, F. Jelezko, R. Laflamme, Y. Nakamura, C. Monroe, and J. L. O'Brien, "Quantum computers," *Nature*, vol. 464, no. 7285. Springer Science and Business Media LLC, pp. 45–53, Mar. 2010. doi: 10.1038/nature08812.
- [3] S. Debnath, N. M. Linke, C. Figgatt, K. A. Landsman, K. Wright, and C. Monroe, "Demonstration of a small programmable quantum computer with atomic qubits," *Nature*, vol. 536, no. 7614. Springer Science and Business Media LLC, pp. 63–66, Aug. 03, 2016. doi: 10.1038/nature18648.
- [4] G. N. Nop, D. Paudyal, and J. D. H. Smith, "Ytterbium ion trap quantum computing: The current state-of-the-art," *AVS Quantum Science*, vol. 3, no. 4. American Vacuum Society, p. 044101, Dec. 2021. doi: 10.1116/5.0065951.
- [5] N. Yazdanpanah and M. K. Tavassoly, "Dipole–dipole interaction between trapped two-level ions interacting with a quantized field in the Lamb–Dicke regime," *Journal of the Optical Society of America B*, vol. 33, no. 3. The Optical Society, p. 382, Feb. 18, 2016. doi: 10.1364/josab.33.000382.
- [6] N. Trautmann and P. Hauke, "Trapped-ion quantum simulation of excitation transport: disordered, noisy, and long-range connected quantum networks," *arXiv*, 2017, doi: 10.48550/ARXIV.1710.09408.
- [7] T. A. Zheng et al., "Measurement of the electric dipole moment of YB171 atoms in an optical dipole trap," *Physical Review Letters*, vol. 129, no. 8, Aug. 2022, doi: 10.1103/physrevlett.129.083001.
- [8] L.-L. Yan, L.-Y. Wang, S.-L. Su, F. Zhou, and M. Feng, "Verification of Information Thermodynamics in a Trapped Ion System," *Entropy*, vol. 24, no. 6. MDPI AG, p. 813, Jun. 11, 2022. doi: 10.3390/e24060813.
- [9] I. Pogorelov et al., "Compact Ion-Trap Quantum Computing Demonstrator," *PRX Quantum*, vol. 2, no. 2. American Physical Society (APS), Jun. 17, 2021. doi: 10.1103/prxquantum.2.020343.
- [10] Bravyi, S., Cross, A.W., Gambetta, J.M. et al. High-threshold and low-overhead fault-tolerant quantum memory. *Nature* 627, 778–782 (2024). <https://doi.org/10.1038/s41586-024-07107-7>

- [11] G. Kirchmair, J. Benhelm, F. Zähringer, R. Gerritsma, C. F. Roos, and R. Blatt, “High fidelity entanglement of $^{43}\text{Ca}^+$ hyperfine clock states,” arXiv, 2008, doi: 10.48550/ARXIV.0812.1976.
- [12] R. Blatt, H. Haffner, C. F. Roos, C. Becher, and F. Schmidt-Kaler, “Ion Trap Quantum Computing with Ca^+ Ions,” Quantum Information Processing, vol. 3, no. 1–5. Springer Science and Business Media LLC, pp. 61–73, Oct. 2004. doi: 10.1007/s11128-004-3105-1.
- [13] P. Wang et al., “Single ion qubit with estimated coherence time exceeding one hour,” Nature Communications, vol. 12, no. 1. Springer Science and Business Media LLC, Jan. 11, 2021. doi: 10.1038/s41467-020-20330-w.